Marques *et al. BMC Pregnancy and Childbirth* (2026) 26:103
https://doi.org/10.1186/s12884-025-08427-w

BMC Pregnancy and Childbirth

RESEARCH Open Access

Factors associated with stillbirth in São Paulo, Brazil: FetRiskS - a multidisciplinary case-control study

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Abstract

Background Stillbirth remains a significant public health issue, with socioeconomic, demographic, environmental, behavioral, healthcare, and biological factors influencing its occurrence. This study aimed to investigate the risk factors for stillbirths in São Paulo, Brazil.

Methods This population-based case-control study included stillbirths (cases) and live births (controls) from 14 public hospitals in the city of São Paulo, Brazil. Data were obtained from interviews with women after delivery, hospital records, antenatal care registries, placental samples, and maternal blood samples for analysis of infection biomarkers. A logistic regression model was used to estimate the odds ratio (OR) with a 95% confidence interval.

Results The study included 401 stillbirths (348 antepartum and 53 intrapartum) and 419 live births. The final model identified significant risk factors for stillbirth, including the absence of a partner (OR = 1.76; 95%CI: 1.15–2.68), smoking during pregnancy (OR = 1.94; 95%CI: 1.10–3.47), inadequate antenatal care access (OR = 2.15; 95%CI: 1.38–3.39), hyperglycemia (OR = 1.91; 95%CI: 1.21–3.03), the presence of placental lesions (OR = 3.15; 95%CI: 2.20–4.52), fetal growth restriction (OR = 6.16; 95%CI: 3.76–10.4), and congenital malformations (OR = 4.25; 95%CI: 2.42–7.67).

Conclusions This study identifies critical maternal and fetal factors associated with stillbirth risk in a middle-income country, showing its complex profile.

Keywords Stillbirths, Fetal mortality, Risk factors, Pregnancy, Case-control

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Introduction

Stillbirth remains a significant public health challenge, affecting many pregnancies globally each year, especially in low- and middle-income countries [1]. Although the global stillbirth rate (SBR) in late pregnancy decreased by 39% between 2000 and 2023, from 22.6 to 14.3 per 1,000 births, an estimated 1.9 million stillbirths still occur annually, with persistent regional disparities driven by several contributing factors [2, 3]. In Brazil, progress has also been made, with the SBR declining by 36.1%, from 9.7 to 6.2[3]. The municipality of São Paulo saw an even greater decrease, with a 44.3% reduction, from 7.9 to 4.4, during the same period [4]. Understanding the causal mechanisms and integrating social-demographic and biological factors are key to reducing fetal mortality [5].

Stillbirths have been described as resulting from multiple pathways, including socioeconomic conditions, environmental exposures, inadequate healthcare access, maternal complications and diseases, behavioral factors, placental alterations, fetal abnormalities, or other disorders of pregnancy [6–9]. While many of these factors are acknowledged, their impact remains insufficiently quantified in low- and middle-income countries, where most stillbirths occur and prevention efforts are needed.

Antenatal care is essential for stillbirth prevention, but evidence remains limited [10]. Early and consistent antenatal care allows for the detection and management of complications, such as hypertensive syndrome, diabetes mellitus, fetal growth restriction (FGR), poor nutrition, and infections. It also helps identify high-risk pregnancies, monitor fetal well-being, and guide timely interventions [10–13]. These conditions, particularly infections, which account for an estimated 10–25% of stillbirths [14–18], and FGR, often associated with placental insufficiency, are among the leading contributors to stillbirths. By compromising oxygen and nutrient delivery to the fetus, placental dysfunction underscores the importance of placental assessment in stillbirth investigations [9, 17, 19]. Access to antenatal care, however, is affected by socioeconomic and demographic factors, including low income, racial disparities, unplanned pregnancies, psychological distress, and lack of social support [20–26]. Beyond these aspects, environmental exposures — such as air pollution — have also been linked to increased stillbirth risk [27–29], highlighting the multifactorial nature of its etiology and the need for integrated prevention strategies.

Many stillbirths can be prevented through high-quality care during pregnancy and delivery [30]. Nevertheless, the pathways from clinical and pregnancy conditions to stillbirth are poorly understood, making it difficult to screen women at increased risk for interventions. This research was conducted in São Paulo, the largest city in Latin America, with broad access to free public health services. Administrative data indicate that 98% of all births occur in hospitals and receive some antenatal care [31]. The majority of stillbirths (82%) occur before birth, and 68% are preterm, similar to the profile of high-income countries [32–34]. We analyzed a broad range of variables to provide a comprehensive understanding of the factors potentially associated with stillbirth. Therefore, this study aimed to investigate the complex interplay of risk factors for stillbirth in the municipality of São Paulo, Brazil.

Methods

Study design and population

This is a population-based case-control study (FetRiskS), in which stillbirths (cases) and live births (controls) were prospectively enrolled at the time of delivery in 14 public hospitals in the municipality of São Paulo (MSP). Of the 14 maternity hospitals included in our study, 9 (64%) are officially registered as reference centers for high-risk pregnancy care. During the data collection period, these hospitals averaged approximately 3,000 live births per year, ranging from 573 (Hospital das Clínicas) to 5,108 (Campo Limpo) in 2020. The inclusion period for participants was from December 2019 to December 2023. During this period, the study was interrupted due to the COVID-19 pandemic, which affected the recruitment and data collection.

The municipality of São Paulo, located in Brazil's Southeast Region, is the country's most developed city. Covering a territory of 1,521 km², with approximately 11,5 million residents. The average per capita income is USD 1,249, corresponding to 4.4 times the Brazilian minimum wage [33]. The MSP has a Human Development Index (HDI) of 0.806 [33]. São Paulo city is a hub for hospitals and health services within the Metropolitan Region of São Paulo (MRSP), which is a highly urbanized and densely populated region with a similar HDI of 0.842 [33].

The sample size was established to compare exposures in a 1:1 unmatched case-to-control ratio with a 95% confidence level (1-α), aiming to detect odds ratios of 1.70, with a test power of 80% (1-β), assuming a 10% exposure for the controls. Using the procedure described by Fleiss [34] with continuity correction, the calculated sample size was 395 cases and 395 controls. To account for potential losses and exclusions (estimated at 5%), a total of 415 stillbirths and 415 live births were considered.

The FetRiskS study included 401 stillbirths and 419 live births. The controls were selected as the subsequent live births after a stillbirth in the same hospital. The unmatched design was chosen to allow assessment of factors that might otherwise be controlled through matching. An additional eight live births were oversampled to account for extreme preterm controls, addressing

gestational age differences between stillbirths and live births. The final sample of cases and controls was specifically designed to be representative of the target population.

Inclusion and exclusion criteria

Women who experienced stillbirths were invited to participate in the hospital study at the participating hospitals. In Brazil, stillbirths are classified at or beyond 22 weeks of gestation or, if the gestational age is unknown, with a birth weight of at least 500 g. Stillbirth was defined as the delivery of a fetus without any signs of life, including the absence of breathing, heartbeat, or movement [35]. This study included only singleton pregnancies of women residing in the MRSP, covering stillbirths and live births between 22 and 44 weeks of gestation or weighing at least 500 g. The study excluded women with communication difficulties (deafness, muteness, psychological/psychiatric disorders, foreign language, etc.) and those whose delivery resulted from court-ordered pregnancy termination.

Data sources

The data source consisted of information from face-to-face interviews with women after delivery, data abstraction from hospital records, and antenatal care registries. The field team consisted of female professionals, all of whom held a college degree and had prior experience collecting data in population surveys. The collectors received specific training for the study and were supervised by a more experienced field researcher, who was responsible for coordinating the activities and providing ongoing support to the team. After collection, the data obtained by all interviewers were systematically verified across each source. Inconsistencies were checked, implausible values were identified, and any discrepancies were resolved when the same information was present in more than one source, ensuring the quality and reliability of the data analyzed.

Maternal blood samples were also analyzed for specific infectious diseases, using either ELISA (enzyme-linked immunosorbent assay) or genome detection by PCR or RT-PCR (reverse transcription polymerase chain reaction). Placental samples were fixed and transported in a 10% buffered formalin solution, followed by fixation for 24 h and examination. The histopathological analysis was performed by a pathologist following a research protocol specifically developed for the study. Environmental data were obtained through secondary sources from the air quality monitoring stations of the Environmental Protection Agency of São Paulo State (CETESB), considering the nearest station to the participants' residences. Additional details can be found in the research protocol [36].

Outcome and exposure variables

The outcome of this study was stillbirth. We described stillbirth and live birth profiles by gestational age and birth weight. Stillbirth was also described as antepartum and intrapartum based on information available in hospital records and maternal interviews, considering factors such as fetal heartbeat at hospital admission, death occurrence in the delivery room, and the moment when the doctor informed the mother about the baby's death. An obstetrician from the research team reviewed this information to classify stillbirth as antepartum or intrapartum.

The exposures of interest were selected based on known or suspected available potential factors associated with stillbirths according to the literature and the author's expertise. The criteria and definitions for the exposure variables are described in Table 1. The risk factor blocks were as follows:

1. Distal factors (Block I):
   Environmental: Air pollution measures through median fine particulate matter concentration (PM2.5 levels in μg/m^3) in the first trimester of pregnancy.
   Socioeconomic factors: Mother's years of schooling (<8 years, ≥8 years), marital status (with partner, without partner), race/ethnicity (white/yellow, brown, black/indigenous), work during pregnancy (no, yes), household income per capita (≤1 and >1 Brazilian minimum wage) and density per room (<3 persons, ≥3 persons).
2. Intermediate factors (Block II):
   Maternal characteristics: Maternal age (<35 years, ≥35 years), adverse perinatal outcomes in previous pregnancies (never got pregnant, multiparous without perinatal adverse outcomes, multiparous with perinatal adverse outcomes), pregestational obesity (no, yes).
   Maternal behaviors: Smoking during pregnancy (no, yes), alcohol and/or drugs during pregnancy (no, yes), adverse maternal and/or family reactions to pregnancy (no, yes).
   Maternal comorbidities: Pregnancy infection (no, yes), hyperglycemia during pregnancy (no, yes), and hypertension during pregnancy (no, yes).
   Pregnancy health care: Antenatal care access (adequate, inadequate, no antenatal care, unknown number of appointments), mother informed of high-risk pregnancy (no, yes), and hospitalization during pregnancy (no, yes).
3. Proximal factors (Block III):
   Placenta: presence of placental lesions (no, yes).
   Fetal attributes: fetal growth restriction (no, yes), congenital malformation (no, yes).

Additionally, we also included the sex of the baby (female, male, undefined/ambiguous/missing), considered independently from the hierarchical structure.

Statistical analysis

Categorical data were described as absolute (n) and relative frequencies (%). Descriptive statistical analysis was performed for each study covariate. The risk of stillbirth was estimated via the odds ratio (OR) with 95% confidence intervals and a significance level of 5% in the Wald Z statistic through a logistic regression model. Covariates were added to the models through the stepwise forward approach.

The variables were entered in blocks from distal to proximal levels, following a hierarchical approach [39]. Initially, the variables from level I (Distal factors), including socioeconomic and environmental factors were inserted, followed by the variables from block II (Intermediate factors), including maternal characteristics, behaviors, comorbidities, and Pregnancy health care, and finally, from block III (Proximal factors), including placentals and fetal attributes, thus forming the final model (I + II + III). All variables were initially included in the model and retained in subsequent steps, regardless of statistical significance, to assess the potential influence of mediating and/or confounding variables. Supplementary Table 1 describes the interpretation of the effects of variables in each model. The proposed causal pathway, prior literature, and biological plausibility guided the ordering of blocks and variables. By adjusting for relevant exposures at the same or earlier levels of the hierarchy, we sought to isolate the independent effects of the primary variables of interest more effectively, considering their clinical and epidemiological importance. To assess multicollinearity, we calculated the generalized variance inflation factor (VIF) [40], with values of 2.5 or greater generally considered indicative of collinearity [41].

The goodness of fit of the final model was evaluated via the deviance statistic (−2 times the log-likelihood). The contribution of each set of variables was evaluated as the difference between the deviances. The statistical significance of differences in deviance was assessed using the chi-square test. The number of degrees of freedom was defined as the difference between the number of parameters in the two models being compared, under the null hypothesis that the model change was significant ($p < 0.05$). Moreover, the Akaike information criterion (AIC) and Bayesian information criterion (BIC) were also used, and smaller values indicated the best fit [42].

Ethics approval and consent to participate

The FetRiskS study was approved by the Ethics Committee of the Municipal Health Department (process n° 16509319.0.3012.5551) and of the Hospital das Clínicas, Faculty of Medicine, University of São Paulo HCFMUSP (process n° 16509319.0.0000.0068). Informed consent was obtained from all pregnant women.

Results

This study included 401 stillbirths and 419 live births. Among the total stillbirths, 348 (86.8%) were antepartum, and 53 (13.2%) were intrapartum (data not shown). The majority (79.3%) of stillbirths were preterm, with 30.2% occurring between 32 and 36 weeks (moderate to late preterm). In contrast, 85.2% of the live births were full-term (37–41 weeks). The proportion of stillbirths decreased as birth weight increased, with 51.6% classified as very low birth weight (< 1,500 g). Moreover, 86.2% of the controls (live births) weighed between 2,500 g and 3,999 g, which aligns with the observed gestational age distribution (Fig. 1).

On the analysis of distal risk factors, mothers of stillbirths were more frequently single (28.9%). They had a threefold greater proportion of residences in households with a high density of people per room (3.7%). Among the intermediate-level variables, mothers with stillbirths were more likely to be ≥35 years of age (25.4%), multiparous with adverse perinatal outcomes (26.2%), and smokers during pregnancy (15.5%). Inadequate (21.9%) or absent antenatal care (4.2%) was also more common among mothers with stillbirths, as was the report of receiving information about high-risk pregnancies from health professionals (51.9%). Additionally, hypertensive syndrome (20.2%) and infections during pregnancy (34.4%) were more frequent among mothers of stillbirths. A higher proportion of mothers of stillbirths had placental injury (63.6%), and a higher proportion of stillbirths had fetal growth restriction (32.4%) and congenital malformations (20.0%) (Table 2).

In the univariate analysis of selected exposures, absence of a partner (OR = 1.62; 95%CI: 1.18,2.24), living in a household with three or more persons per room (OR = 3.25; 95%CI: 1.25,10.1), mothers with a previous adverse perinatal outcome (OR = 1.77; 95%CI: 1.24,2.54), smoking during pregnancy (OR = 2.21; 95%CI: 1.42,3.51), inadequate (OR = 1.86; 95%CI: 1.29,2.69) or absent antenatal care (OR = 10.4; 95%CI: 2.95,66.0), infection during pregnancy (OR = 1.43; 95%CI: 1.06,1.93), hypertension (OR = 2.13; 95%CI: 1.44,3.18), and mother informed of high-risk pregnancy (OR = 1.71; 95%CI: 1.29,2.27) were associated with a higher risk of stillbirth. Additionally, fetuses with placental lesions (OR = 3.43; 95%CI: 2.55,4.64), fetal growth restriction (OR = 7.25; 95%CI: 4.70,11.6), congenital malformation (OR = 3.77; 95%CI: 2.39,6.11), and undefined/ambiguous/absent fetal sex (OR = 5.03; 95%CI: 1.30,33.0) also showed an increased risk of stillbirth (Table 2).

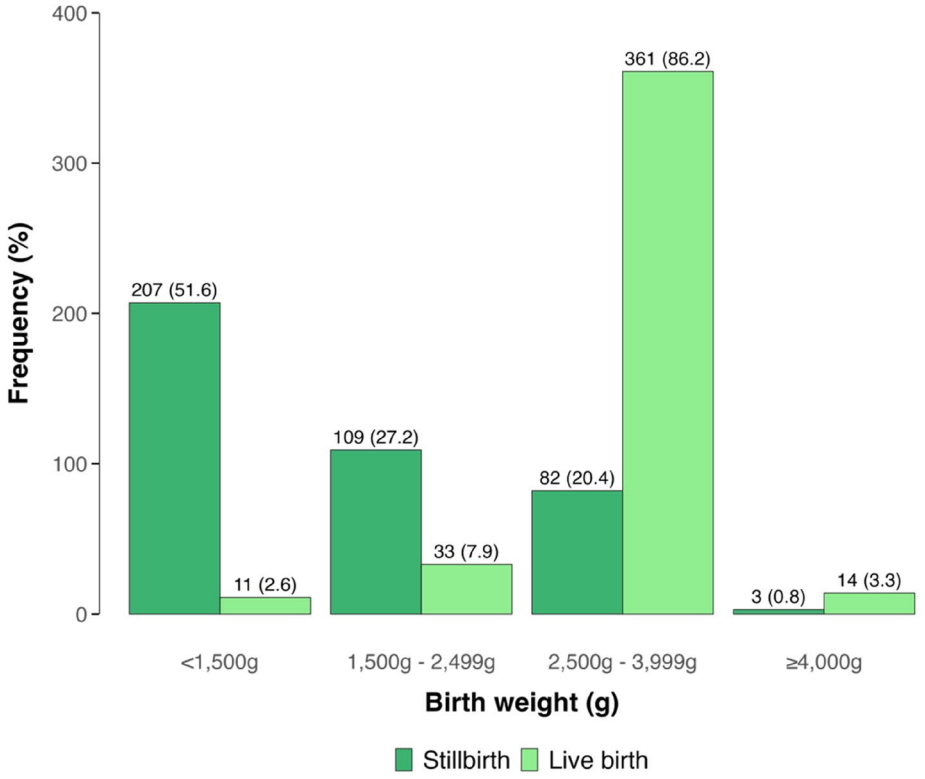
Table 1 Definition of study variables

Variable	Source	Categories	Definition criteria
Air pollution (PM2.5)	Environmental Agency of São Paulo (CETESB)	Median particulate matter concentration (PM2.5) in the first trimester of pregnancy	Based on PM2.5 (μg/m^3) measurements from the monitoring station nearest to the mother's residence
Mother's years of schooling	Interview with the mother	< 8 years; ≥ 8 years	Reported by the woman in the interview
Marital status		With partner; Without partner	
Race/ethnicity		White/yellow; Brown; Black/indigenous	
Worked during pregnancy		No; yes	
Household income per capita		≤ 1 Brazilian minimum wage; >1 Brazilian minimum wage	
Density per room		< 3 persons; ≥ 3 persons	
Maternal age		< 35 years; ≥ 35 years	
Adverse perinatal outcomes in previous pregnancies		Never got pregnant; Multiparous without perinatal adverse outcomes; Multiparous with perinatal adverse outcomes	Adverse perinatal outcomes included stillbirth, low birth weight, and preterm birth.
Pregestational obesity	Antenatal care records, interview with the mother	No; Yes	Obesity was defined as BMI ≥ 30 kg/m^2. Pre-pregnancy weight data were obtained from antenatal care records or self-reported by the participant during the hospital interview.
Smoking during pregnancy	Interview with the mother	No; Yes	Reported by the woman in the interview
Alcohol and/or drugs during pregnancy		No; Yes	
Adverse maternal and/or family reaction to pregnancy		No; Yes	Based on the woman's report of her own and her family's reaction. An adverse reaction was classified as 'yes' when the response was 'Indifferent', 'Upset', or 'Doesn't remember/Doesn't want to answer', and as 'no' when the response was 'Happy'.
Antenatal care access	Interview with the mother	Adequate; Inadequate; No antenatal care; With antenatal care, unknown number of appointments	The "no antenatal care" category includes women who did not have access to antenatal care. Access to antenatal care was defined as inadequate if at least one of the following situations occurred: - First visit after the first trimester of pregnancy - Gestational age at birth < 24 weeks, and attended less than 3 visits - Gestational age at birth ≥ 24 and < 28 weeks, and attended less than 4 visits - Gestational age at birth ≥ 28 and < 32 weeks, and attended less than 5 visits - Gestational age at birth ≥ 32 weeks, and attended less than 6 visits. The category "With antenatal care, unknown number of appointments" corresponds to women who had access to antenatal care but information was not obtained on the number of visits and/or the initiation of antenatal care.
Pregnancy infection	Maternal blood tests at the time of delivery	No; Yes	Defined as reactive IgM immunoglobulin in maternal blood, measured by ELISA or genome detection by PCR or RT-PCR, also in maternal blood, for the agents: Parvovirus B19, Cytomegalovirus, Treponema pallidum, *Chlamydia trachomatis*, *Toxoplasma gondii*, dengue virus and Zika virus. For *Streptococcus agalactiae* and Human Herpesvirus 6, only PCR or RT-PCR were performed due to the absence of commercial serological tests in Brazil.

Table 1 (continued)

Variable	Source	Categories	Definition criteria
Hyperglycemia	Antenatal care record, Hospital records, Interview with the mother	No; Yes	For women who had an antenatal care record, a presence of hyperglycemia was considered if they have at least one fasting glycemia test ≥ 92 mg% during pregnancy; the diagnosis of gestational diabetes was also considered. For women without an antenatal care record, the diagnosis of diabetes during pregnancy recorded in the hospital record was considered. For women without both records, self-reported diagnosis of diabetes during pregnancy was considered.
Hypertension	Antenatal care record, Hospital records, Interview with the mother	No; Yes	For women who had an antenatal care record, a presence of hypertension was considered if they have, in the second and/or third pregnancy trimester, at least two pressure measurements with systolic pressure ≥ 140 mmHg and/or diastolic pressure ≥ 90 mmHg. For women without an antenatal care record, the diagnosis of high blood pressure during pregnancy recorded in the hospital record was considered. For women without both records, self-reported diagnosis of high blood pressure, in the second and/or third trimester, was considered.
Mother informed of high-risk pregnancy	Interview with the mother	No; Yes	Reported by the woman in the interview
Hospitalization during pregnancy			
Placental lesions	Pathological report of the placenta	No; Yes	The presence of at least one placental lesion was considered. Placental lesions were defined according to the Amsterdam international consensus criteria for placental diagnosis [37], considering the presence of **inflammatory signs** (chorioamnionitis, omphalitis, villitis, intervillositis, or deciduitis), **maternal vascular malperfusion** (villous infarction, hematoma, hypertrophic decidual vasculopathy, or decidual vasculopathy with acute atherosis), and **fetal vascular malperfusion** (thrombosis, groups of avascular villi, hemorrhagic endovasculitis, or umbilical vessel thrombosis).
Fetal growth restriction	Antenatal care records, Hospital records, Interview with the mother, Delivery room book	No; Yes	The parameters defined by Intergrowth-21st were adopted to classify fetal growth according to birth weight for gestational age and sex. SGA, defined as below p10, was classified as a proxy for FGR. For this classification, we used the gigs package in R, which provides INTERGROWTH-21st references for 22 to 44 weeks of gestation [38].
Congenital malformation	Hospital records, Interview with the mother	No; Yes	The presence of congenital malformation was considered through identification in antenatal ultrasound, detection of the presence of congenital malformation after birth or the mother's report in the interview.
Birth weight (grams)	Delivery room book	< 1,500, 1,500-2,499, 2,500-3,999, ≥ 4,000	Birth weight obtained in grams
Gestational age (weeks)	Antenatal care records, Hospital records, Interview with the mother, Delivery room book	22-27, 28–31, 32–36, 37–41, 42-44	Gestational age (GA) was considered in completed weeks. Derived by developing an algorithm that incorporated the calculation of GA through: 1. Last Menstrual Period (LMP) ± 5 days difference from Ultrasound (US) ≤ 13 weeks; 2. US ≤ 13 weeks; 3. LMP ± 10 days difference from US 14–19 weeks; 4. US 14–19 weeks; 5. GA recorded in the delivery room book; 6. LMP based on maternal interview.
Fetal sex	Hospital records	Female; Male; Undefined/ambiguous/unknown	The classification was abstracted as it appeared in the hospital records.

CETESB Environmental Agency of São Paulo, *BMI* Body mass index, *ELISA* Enzyme-linked immunosorbent assay, *RT-PCR* Reverse transcription polymerase chain reaction, *FGR* Fetal growth restriction, *SGA* Small for gestational age, *GA* Gestational age, *LMP* Last menstrual period, *US* Ultrasound



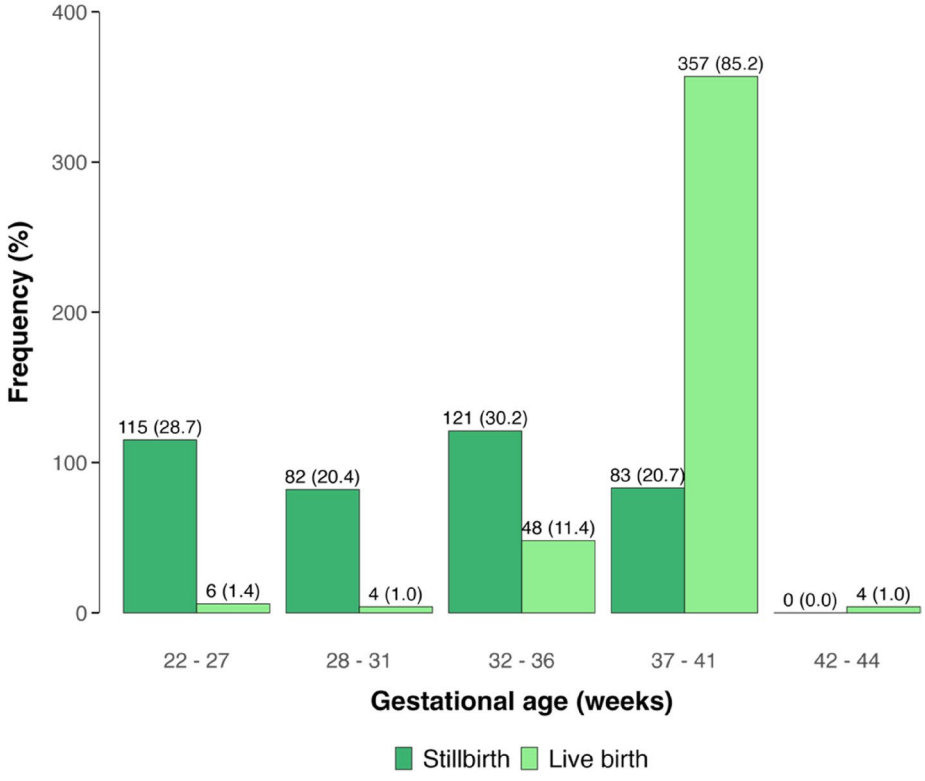


Fig. 1 Distribution of stillbirths and live births by gestational age and birth weight, municipality of São Paulo – Brazil, 2019–2023

Table 2 Univariate analysis of risk factors associated with stillbirth, municipality of São Paulo – Brazil, 2019–2023

Variables	Stillbirths *N* = 401		Live births *N* = 419		Crude OR	95%CI	*p* value
	N	%	*N*	%			
Distal level							
Mother's years of schooling							
≥ 8 years	345	86.0	364	86.9	Ref.		
< 8 years	54	13.5	51	12.2	1.12	0.74, 1.69	0.600
Unknown	2	0.5	4	1.0			
Marital status							
With partner	285	71.1	335	80.0	Ref.		
Without partner	116	28.9	84	20.0	1.62	1.18, 2.24	0.003
Race/ethnicity							
White/Yellow	122	30.4	143	34.1	Ref.		
Brown	210	52.4	209	49.9	1.18	0.87, 1.60	0.300
Black/Indigenous	68	17.0	67	16.0	1.19	0.79, 1.80	0.400
Unknown	1	0.2	-	-			
Worked during gestation							
No	183	45.6	185	44.2	Ref.		
Yes	218	54.4	234	55.8	0.94	0.72, 1.24	0.700
Household income per capita							
> 1 minimum wage	126	31.4	140	33.4	Ref.		
≤ 1 minimum wage	273	68.1	279	66.6	1.09	0.81, 1.46	0.600
Unknown	2	0.5	-	-			
Density per room							
< 3 persons	382	95.3	414	98.8	Ref.		
≥ 3 persons	15	3.7	5	1.2	3.25	1.25, 10.1	0.024
Unknown	4	1.0	-	-			
Intermediate level							
Mother's age							
< 35 years	299	74.6	339	80.9	Ref.		
≥ 35 years	102	25.4	80	19.1	1.45	1.04, 2.02	0.029
Previous adverse perinatal pregnancy outcome							
Never got pregnant	132	32.9	138	32.9	1.21	0.88, 1.66	0.200
Multiparous without perinatal adverse outcomes	163	40.6	206	49.2	Ref.		
Multiparous with perinatal adverse outcomes	105	26.2	75	17.9	1.77	1.24, 2.54	0.002
Unknown	1	0.2	-	-			
Pregestational obesity							
No	269	67.1	295	70.4	Ref.		
Yes	117	29.2	113	27.0	1.14	0.84, 1.54	0.400
Unknown	15	3.7	11	2.6	1.50	0.68, 3.39	0.300
Smoking during pregnancy							
No	339	84.5	387	92.4	Ref.		
Yes	62	15.5	32	7.6	2.21	1.42, 3.51	< 0.001
Alcohol and/or drugs during pregnancy							
No	363	90.5	388	92.6	Ref.		
Yes	38	9.5	31	7.4	1.31	0.80, 2.16	0.300
Negative maternal and/or family reaction to pregnancy							
No	275	68.6	289	69.0	Ref.		
Yes	126	31.4	126	30.1	1.05	0.78, 1.41	0.700
Unknown	-	-	4	1.0			
Antenatal care access							
Adequate	268	66.8	328	78.3	Ref.		
Inadequate	88	21.9	58	13.8	1.86	1.29, 2.69	0.001
No antenatal care	17	4.2	2	0.5	10.4	2.95, 66.0	0.002

Table 2 (continued)

Variables	Stillbirths *N* = 401		Live births *N* = 419		Crude OR	95%CI	*p* value
	N	%	*N*	%			
With antenatal care, unknown n° appointments	28	7.0	31	7.4	1.11	0.64, 1.89	0.700
Infecction during pregnancy							
No	251	62.6	294	70.2	Ref.		
Yes	138	34.4	113	27.0	1.43	1.06, 1.93	0.019
Unknown	12	3.0	12	2.9	1.17	0.51, 2.68	0.700
Hyperglycemia							
No	321	80.0	360	85.9	Ref.		
Yes	74	18.5	58	13.8	1.43	0.98, 2.09	0.061
Unknown	6	1.5	1	0.2			
Hypertension							
No	315	78.6	373	89.0	Ref.		
Yes	81	20.2	45	10.7	2.13	1.44, 3.18	< 0.001
Unknown	5	1.2	1	0.2			
Mother informed of high-risk pregnancy							
No	175	43.6	245	58.5	Ref.		
Yes	208	51.9	170	40.6	1.71	1.29, 2.27	< 0.001
Unknown	18	4.5	4	1.0	6.30	2.30, 22.1	0.001
Hospitalization during pregnancy							
No	340	84.8	361	86.2	Ref.		
Yes	59	14.7	54	12.9	1.16	0.78, 1.73	0.500
Unknown	2	0.5	4	1.0			
Proximal level							
Placental lesions							
No	118	29.4	240	57.3	Ref.		
Yes	255	63.6	151	36.0	3.43	2.55, 4.64	< 0.001
Missing	28	7.0	28	6.7	2.03	1.15, 3.60	0.014
Fetal growth restriction							
No	261	65.1	391	93.3	Ref.		
Yes	130	32.4	26	6.2	7.25	4.70, 11.6	< 0.001
Unknown	10	2.5	2	0.5			
Congenital malformation							
No	321	80.0	393	93.8	Ref.		
Yes	80	20.0	26	6.2	3.77	2.39, 6.11	< 0.001
Fetal sex							
Female	186	46.4	187	44.6	Ref.		
Male	205	51.1	230	54.9	0.90	0.68, 1.18	0.400
Undefined/ambiguous/missing	10	2.5	2	0.5	5.03	1.30, 33.0	0.039

Table 3 displays the results of the models for each block in the hierarchical analyses. The first column represents the distal model (Block I). The second column shows the intermediate model, which combines Block I (distal factors) with Block II (intermediate factors). The third column presents the final model, adding Block III (proximal factors) to the previous two blocks. Owing to its hierarchical nature, the results can be interpreted horizontally (for a given variable across adjustments) and diagonally (adjustment between blocks).

In Model I (distal factors), mothers without a partner faced a 65% greater risk of stillbirth. Living in households with three or more people per room was associated with a 3.27-fold increased risk of stillbirth. However, factors such as exposure to air pollution (PM2.5), the mother's years of schooling, household income per capita, race/ethnicity, and whether the mother worked during gestation had no significant impact on the risk of stillbirth (Table 3).

In Model II (distal and intermediate factors), mothers without a partner and living in households with three or more people per room remained associated with an increased risk of stillbirth. Among intermediate factors, women with previous adverse perinatal outcomes had a 74% higher risk of stillbirth. Inadequate access to antenatal care was associated with a 93% higher risk,

Table 3 Factors associated with stillbirth according to the block of risk factors, municipality of São Paulo – Brazil, 2019–2023

Variables	Model I			Model I + II			Model I + II + III		
	OR[a]	95% CI	p value	OR[a]	95% CI	p value	OR[a]	95% CI	p value
Distal level									
Air pollution (PM 2.5 μg/m^3)	1.00	0.96, 1.04	0.910	0.99	0.96, 1.03	0.800	1.02	0.97, 1.06	0.500
Mother's years of schooling									
≥ 8 years	Ref.			Ref.			Ref.		
<8 years	1.04	0.67, 1.61	0.800	0.95	0.60, 1.51	0.800	0.94	0.56, 1.58	0.800
Marital status									
With partner	Ref.			Ref.			Ref.		
Without partner	1.65	1.18, 2.32	0.003	1.46	1.00, 2.13	0.048	1.76	1.15, 2.68	0.009
Race/ethnicity									
White/Yellow	Ref.			Ref.			Ref.		
Brown	1.24	0.90, 1.70	0.200	1.30	0.93, 1.83	0.130	1.30	0.89, 1.91	0.200
Black/Indigenous	1.17	0.76, 1.80	0.500	1.19	0.75, 1.88	0.500	1.15	0.69, 1.93	0.600
Worked during gestation									
No	Ref.			Ref.			Ref.		
Yes	0.98	0.72, 1.31	0.900	1.00	0.73, 1.37	>0.9	0.83	0.58, 1.18	0.300
Household income per capita									
>1 minimum wage	Ref.			Ref.			Ref.		
≤ 1 minimum wage	0.94	0.68, 1.30	0.700	0.95	0.67, 1.34	0.800	0.97	0.66, 1.44	0.900
Density per room									
< 3 persons per room	Ref.			Ref.			Ref.		
≥ 3 persons per room	3.27	1.23, 10.3	0.025	3.24	1.16, 10.6	0.034	2.69	0.83, 9.97	0.110
Intermediate level									
Mother's age									
< 35 years				Ref.			Ref.		
≥ 35 years				1.38	0.94, 2.01	0.100	1.44	0.94, 2.21	0.100
Previous adverse perinatal outcome									
Never got pregnant				1.40	0.98, 2.02	0.066	1.17	0.78, 1.75	0.500
Multiparous without adverse perinatal outcomes				Ref.			Ref.		
Multiparous with adverse perinatal outcomes				1.74	1.17, 2.60	0.007	1.47	0.94, 2.30	0.091
Pregestational obesity									
No				Ref.			Ref.		
Yes				1.05	0.74, 1.48	0.800	1.14	0.78, 1.68	0.500
Unknown				1.36	0.53, 3.46	0.500	1.92	0.69, 5.34	0.200
Smoking during pregnancy									
No				Ref.			Ref.		
Yes				2.17	1.30, 3.68	0.003	1.94	1.10, 3.47	0.023
Alcohol and/or drugs during pregnancy									
No				Ref.			Ref.		
Yes				1.11	0.63, 1.97	0.700	1.07	0.57, 2.01	0.800
Negative maternal and/or family reaction to pregnancy									
No				Ref.			Ref.		
Yes				0.83	0.59, 1.16	0.300	0.81	0.55, 1.18	0.300
Antenatal care access									
Adequate				Ref.			Ref.		
Inadequate				1.93	1.29, 2.91	0.002	2.15	1.38, 3.39	<0.001
No antenatal care				11.2	0.53, 542.0	0.150	27.1	1.04, 1,541	0.063
With antenatal care, unknown n° appointments				1.01	0.56, 1.81	>0.9	0.81	0.41, 1.58	0.500
Infecction during pregnancy									
No				Ref.			Ref.		
Yes				1.49	1.07, 2.08	0.018	1.39	0.96, 2.02	0.081
Unknown				1.12	0.45, 2.75	0.800	1.09	0.38, 3.05	0.900
Hyperglycemia									

Table 3 (continued)

Variables	Model I			Model I + II			Model I + II + III		
	OR[a]	95% CI	p value	OR[a]	95% CI	p value	OR[a]	95% CI	p value
No				Ref.			Ref.		
Yes				1.44	0.95, 2.18	0.084	1.91	1.21, 3.03	0.006
Hypertension									
No				Ref.			Ref.		
Yes				1.60	1.03, 2.52	0.038	1.43	0.86, 2.38	0.200
Mother informed of high-risk pregnancy									
No				Ref.			Ref.		
Yes				1.58	1.14, 2.18	0.006	1.08	0.73, 1.58	0.700
Unknown				0.98	0.04, 11.2	>0.9	0.46	0.02, 6.49	0.600
Hospitalization during pregnancy									
No				Ref.			Ref.		
Yes				0.88	0.56, 1.36	0.600	0.88	0.53, 1.45	0.600
Proximal level									
Placental lesion									
No							Ref.		
Yes							3.15	2.20, 4.52	<0.001
Missing							1.75	0.90, 3.39	0.100
Fetal growth restriction									
No							Ref.		
Yes							6.16	3.76, 10.4	<0.001
Congenital malformation									
No							Ref.		
Yes							4.25	2.42, 7.67	<0.001
Fetal sex									
Female							Ref.		
Male							0.86	0.61, 1.21	0.400
Undefined/ambiguous/missing							5.25	1.15, 38.0	0.052
Model fitness									
Log-likelihood	−540			−505			−425		
Deviance	1.079			1.010			849		
Residual d.f.	782			764			758		
p value (Deviance Chi-square test)	<0.001			<0.001			<0.001		
Akaike information criterion (AIC)	1.097			1.064			915		
Bayesian information criterion (BIC)	1.139			1.190			1.069		

[a]Adjusted odds ratio

whereas infections during pregnancy increased the risk by 49%. Hypertension increased the risk by 60%, and being informed of a high-risk pregnancy was linked to a 58% higher risk of stillbirth. Additionally, smoking during pregnancy was associated with a 2.17-fold greater risk of stillbirth. No statistically significant associations were found between stillbirth and factors such as maternal age, pregestational obesity, alcohol and/or drug use during pregnancy, negative maternal and/or family reactions to pregnancy, hyperglycemia, or hospitalization during pregnancy (Table 3).

The final model, which includes distal, intermediate, and proximal factors, incorporates all the variables from the three blocks. After adjusting for proximal exposures, only marital status remained significantly associated with stillbirth at the distal level, with mothers without a partner having a 76% higher risk of stillbirth than those with a partner. At the intermediate level, the risk of stillbirth was 94% greater for smokers during pregnancy, 2.15 times greater for those with inadequate access to antenatal care, and 91% greater for those with hyperglycemia. At the proximal level, the presence of placental lesions was associated with a 3.15-fold increased risk, fetal growth restriction with a 6.16-fold increased risk, and congenital malformations with a 4.25-fold increased risk of stillbirth (Table 3; Fig. 2). In our analysis, all VIF values were below 1.5, indicating no relevant multicollinearity (Table S2).

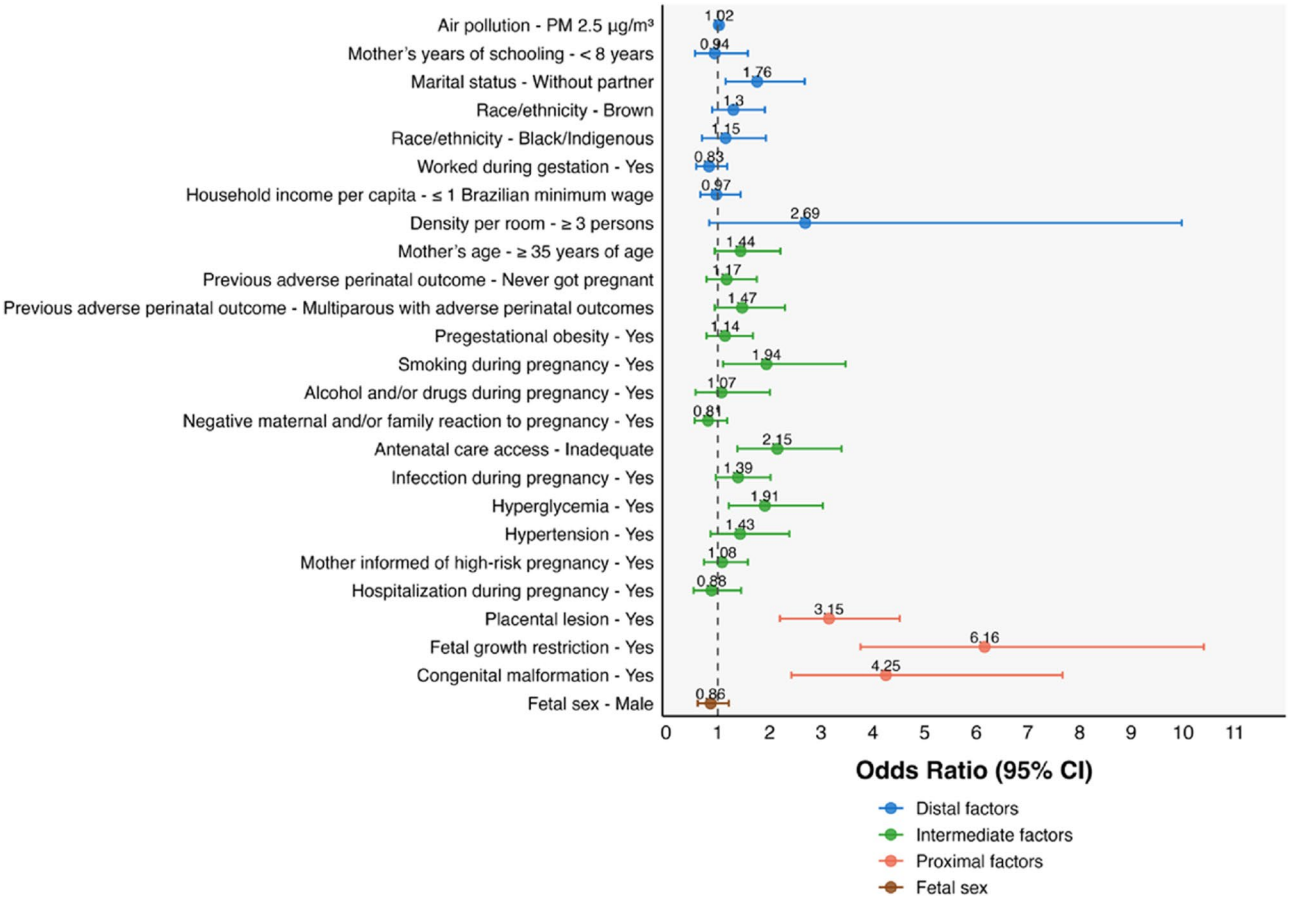


Fig. 2 Forest plot of adjusted odds ratio for stillbirth risk factors, municipality of São Paulo – Brazil, 2019–2023. Note: Only covariate risk categories were included

Discussion

This study provides insights into factors associated with stillbirth risk in São Paulo, Brazil. The final model, which included distal, intermediate, and proximal levels, identified several key risk factors of stillbirth. At the distal level, mothers without a partner and living in overcrowded conditions (three or more persons per room) were associated with stillbirth. Factors at the intermediate level, including being multiparous with previous adverse perinatal outcomes, having infections during pregnancy, having hypertension syndrome, and being informed of a high-risk pregnancy, were associated with stillbirths in earlier models, but these variables were no longer significant in the final model. Ultimately, the final model emphasized mothers without a partner, smoking during pregnancy, inadequate antenatal care access, hyperglycemia, and proximal factors such as the presence of placental lesions, fetal growth restriction, and congenital malformations as strong risk factors of stillbirth.

Most stillbirths were preterm, had very low birth weights, and occurred antepartum. São Paulo is a middle-income city marked by significant social inequality [43]. This result is possibly due to the high rate of antenatal care (96.2% of live births have had 4 or more antenatal visits) and hospital deliveries (98% of all births) found in São Paulo municipality [31]. Additionally, the stillbirth profile is similar to that of developed countries, where antepartum and preterm stillbirths are predominant [32]. Only a small proportion of stillbirths were intrapartum, which is also consistent with trends observed in high-income countries [44–46].

Our study revealed no significant associations between the environmental factor, some socioeconomic factors, and stillbirth risk, such as exposure to air pollution (PM2.5), the mother's years of schooling, household income per capita, race/ethnicity, and whether the mother worked during gestation, despite their well-documented impact on pregnancy outcomes. These factors have been linked to stillbirths and may add layers of vulnerability, limit access to adequate antenatal care, influence maternal behavior, and affect the management of high-risk pregnancies [20, 21, 28], but our results did not find this association. Regarding air pollution, mothers in both groups were exposed to high levels of PM2.5 during pregnancy, exceeding the World Health Organization (WHO) air quality guidelines (< 5 µg/m³ annually

and < 15 μg/m³ for up to 3–4 days/year). In our study, the median pollution levels for both groups during the first trimester were more than twice the WHO-recommended limit [47].

Concerning socioeconomic factors, the lack of association for certain variables can be attributed to the homogeneity observed in the sociodemographic profiles of our cases and controls or to easy access to antenatal care. Our data were derived from users of public hospitals in São Paulo, which are part of the universal Brazilian Unified Health System (SUS). These hospitals primarily serve low-income populations, with public facilities in the MSP accounting for 64% of all deliveries [48]. According to a 2019 health survey, among individuals admitted to public hospitals in MSP, 62% identified as Black, 60% had an income of up to one minimum wage, and 65% had not completed high school [49].

Among the distal factors, the risk of stillbirth increased for mothers without a partner and those living in overcrowded conditions (three or more persons per room). A mother without a partner indicates a possible lack of psychological or financial support. Single-mother families are more likely to experience poverty because of the loss of a partner's finances and discriminatory wages against women [50]. Our findings align with those of a Brazilian study on perinatal mortality, which reported a 2.44 increased risk [51] for mothers without partners. It also aligns with the international literature, such as the Canadian birth census cohort, where single mothers faced a 92% higher risk of stillbirth than married mothers [52]. The risk was also increased for preterm birth (36%) and small for gestational age (SGA) (31%) [53]. Similarly, a retrospective cohort study in South Australia found that unmarried women had a 20% greater risk of stillbirth [52].

Furthermore, high density per room was associated with an increased risk of stillbirth in the first two models, but lost significance in the final model. This variable is recognized as an important, wider factor of health and serves as a measure of household crowding, considering all rooms in a dwelling and the number of occupants. This metric can reflect situations where individuals share living spaces with non-family members, such as in tenements, and may be associated with vulnerable conditions [54]. Research indicates that poor housing conditions, including overcrowding, can cause stress in pregnant women and are associated with adverse maternal and child health outcomes, some of which have long-term effects [55]. While rarely explored in stillbirths, housing has been identified as a social determinant of low birth weight and preterm birth, underscoring the impact of social disparities on fetal development [56].

For intermediate factors, multiparous women with adverse perinatal outcomes, infection during pregnancy, hypertension syndrome, and mothers being informed of a high-risk pregnancy were associated with stillbirth, but their significance diminished when proximal variables were included. However, smoking during pregnancy, hyperglycemia, and inadequate antenatal care remained significant, indicating an increased risk for stillbirths in the final model.

Smoking during pregnancy was associated with an increased risk of stillbirth in the presence of distal and intermediate variables. In our study, smoking during pregnancy doubled the risk of stillbirth. A similar result was reported in a cohort of thirteen UK maternity units, where smoking increased the risk of preterm stillbirths by 70% [43]. Smoking is a well-recognized modifiable risk factor that induces oxidative stress, which affects the expression of key transcription factors involved in angiogenesis. This leads to placental dysfunction, contributing to several adverse pregnancy outcomes, including low birth weight, preterm birth, fetal growth restriction, antepartum and intrapartum stillbirth, and infant morbidity and mortality [57–59]. Research has shown that pregnant women who smoke are more likely to have lower levels of education [60], lower monthly family income, and less regular follow-up with antenatal care visits than women who are exposed to passive smoking or those who do not smoke [61].

In our study, mothers with hyperglycemia had nearly double the risk of stillbirth, which aligns with findings that raised blood glucose levels more than tripled the odds of stillbirth and perinatal mortality [62]. The effect of hyperglycemia is evident through proximal-level variables, including placental lesions and fetal growth restriction. Maternal hyperglycemia, a common pregnancy complication, is associated with adverse outcomes such as macrosomia, fetal growth restriction [63, 64], preterm birth, birth injury, perinatal morbidity, and mortality [66, 67]. It also leads to functional and structural placental abnormalities, including villous maturation defects [65, 66]. A national population-based retrospective cohort study revealed that pregnancies affected by preexisting diabetes or gestational diabetes, particularly those with abnormal fetal growth, have an elevated risk of stillbirth as gestational age progresses [64].

Inadequate access to antenatal care was previously associated with stillbirth, and this association remained significant after adjusting for strong proximal risk factors. Our findings align with results from a retrospective cohort study in South Australia, which identified that inadequate access to antenatal care was associated with a fourfold increased risk of stillbirth [52]. Additionally, a previous study showed that the quality of antenatal care decreases the odds of antepartum stillbirth by almost 81% after controlling for other factors [68]. Women with late or no antenatal care have a greater risk of stillbirth

and obstetric complications owing to inadequate support, whereas adequate antenatal care helps prevent these outcomes by identifying risks, managing morbidities, promoting healthy behaviors, ensuring access to treatment, and benefiting women with chronic conditions and pregnancy-related complications [10].

The proximal level included factors related to placental lesions, fetal growth restriction (FGR), and congenital malformation, all of which are associated with an increased risk of stillbirth after adjustment for other variables. Placental abnormalities are well-known risk factors for stillbirth, and their examination is a key tool for understanding both maternal and fetal etiologies [9]. Approximately 93% of stillbirths and live births in our sample underwent placental pathology exams, aiding in the identification of placental abnormalities and other etiologies. In our study, placental lesions were defined by the presence of at least one alteration, including inflammatory signs, maternal vascular malperfusion (MVM), or fetal vascular malperfusion (FVM). The results from the Stillbirth Collaborative Research Network (SCRN), a U.S. case-control study, revealed that MVM and FVM lesions were more prevalent in stillborn placentas than in both term and preterm live births, occurring in 81.3% and 85.2% of cases, respectively. Placentas with MVM and FVM are associated with 48% and 34% greater risks of stillbirth, respectively [12]. Additionally, MVM is the most common placental disease contributing to FGR [69, 70].

Fetal growth restriction was associated with an increased risk of stillbirth at the proximal level, even after adjusting for confounders. It emerged as one of the highest risks in the study compared with other levels in the final model. Our results are consistent with those of previous studies, such as the Intergrowth-21st study, which revealed that FGR was associated with a 4.6-fold greater risk of antepartum stillbirth [71]. Additionally, the prevalence of FGR in our study (32% in stillbirths and 6% in live births) is consistent with findings from a population-based birth cohort study conducted in São Paulo City between 2017 and 2019, which reported rates of 36% and 6%, respectively [48]. FGR remains the strongest and most common underlying cause of preventable stillbirth [18, 69]. Its causes are diverse, primarily stemming from placental dysfunction and abnormal development [69, 70]. However, FGR has a multifactorial etiology involving maternal conditions (e.g., hypertensive disorders, diabetes), behavioral factors (e.g., smoking, malnutrition), congenital infections (e.g., cytomegalovirus and syphilis), and fetal disorders (e.g., genetic or structural anomalies) [72]. Moreover, a limited understanding of the mechanisms regulating fetal growth pathways poses a challenge for implementing more effective interventions [73].

Our findings indicate that congenital malformations are an independent risk factor for stillbirth, even after adjustment for covariates. This finding is consistent with the results of the SCRN, which reported a similar risk to that reported in our study, with a 4.3-fold increase in stillbirth among pregnancies with fetal anomalies. Additionally, the risk increased with the number of detected anomalies. The prevalence of congenital malformations in our study was also comparable, with 20.0% for stillbirths and 6.2% for live births, whereas SCRN revealed major anomalies in 23.4% of stillbirths compared with 4.3% of live births [74]. Compared with other stillbirth risk factors, guidelines for managing pregnancies affected by isolated fetal anomalies are scarce. In addition to stillbirth risk, fetuses with some chromosomal abnormalities and/or genetic syndromes are also at risk for a range of complications, which reflects the potential for medical intervention and impacts perinatal morbidity and survival [72].

In the final model, the loss of statistical significance of living in overcrowded conditions, previous adverse perinatal outcomes, infection during pregnancy, hypertension syndrome, and mothers being informed of a high-risk pregnancy could indicate that their effects on stillbirth are mediated by proximal exposures [39]. Further analyses of our data are necessary to analyze the mediating effects on stillbirth.

Strengths and limitations

A key strength of this study is its pioneering approach, which uses multiple data sources to comprehensively assess stillbirth risk factors in a population-based design. Our estimates are trengths and limitationss are essential, especially in Brazil, owing to the characteristics of inequality, inequity, and environmental context, which result in distinct distal and intermediate risk factors. This contextual knowledge contributes to a more effective implementation of public policies. The use of multiple data sources and the inclusion of biological materials, which are rarely present in population-based studies, are significant strengths of our study. We obtained placental data for the vast majority of cases and controls and conducted multiple tests on maternal blood samples to detect infections associated with stillbirth risk. This research represents the first of many investigations that will emerge from the FetRiskS study, providing a foundation for further analyses of the complex risk factors of stillbirth.

Our data were collected from public hospitals, primarily located in the periphery and poorer areas of São Paulo city. This approach was chosen strategically, as conducting interviews and collecting biological samples in private hospitals is particularly challenging. However,

as mentioned earlier, this choice resulted in a relatively homogeneous socioeconomic profile, which may not fully reflect the socioeconomic disparities across São Paulo.

Our study revealed maternal characteristics and behaviors that are not available in publicly accessible data sources. The investigation of these factors in a population-based study is unprecedented in the municipality of São Paulo. The identification of these aspects contributes to the development of strategies and health policies aimed at improving the quality of women's healthcare and access to antenatal care, which plays a crucial role in the early detection of conditions, as well as in the guidance and treatment of maternal diseases. However, we acknowledge the potential for self-reporting bias regarding behavioral variables, as these were collected through interviews conducted after delivery, and the willingness to respond to sensitive questions may have differed between cases and controls.

Biases related to data extracted from hospital records and antenatal care cards should also be considered, as these sources may be incomplete or contain recording errors. To minimize the impact of missing data, the classification of certain exposures was based on information available from multiple data sources rather than relying on a single variable to reduce the occurrence of missing entries. Additionally, recall bias is a potential limitation for the information obtained through interviews with mothers, as memory lapses or inaccuracies could influence the reliability of the data. Furthermore, grieving the loss of their fetus may have led participants to be reluctant to disclose behaviors perceived negatively, such as alcohol or drug use.

The hypertension and hyperglycemia variables were created by combining data from multiple sources to reduce missing data. Both variables serve as screening markers for hyperglycemia and hypertension at some point during pregnancy. In addition, the air pollution variable was a contextual variable obtained from air quality measurements at the station nearest the mother's residence. This approach does not represent an individual direct measure, which may have influenced the results, as the mothers' residences were relatively concentrated in the same areas of the city. Furthermore, it was not possible to observe a statistically significant association of some known risk factors with fetal mortality. Given the small number of live births without antenatal care access, the confidence intervals were wide, potentially limiting the interpretation of the magnitude of the effect size.

Conclusions

The analysis of risk factors across different levels revealed a complex interplay of socioeconomic, health care, maternal, and fetal factors contributing to stillbirth risk. These findings emphasize the multifactorial nature of stillbirth and underscore the need for targeted interventions to improve antenatal care and mitigate modifiable risk factors. Strategies should include support for single mothers and expanded access to antenatal care for the early identification and proper management of maternal conditions, such as hyperglycemia. Additionally, guidance on harmful behaviors, such as smoking during pregnancy, is crucial. Continuous monitoring of fetal growth, along with the identification of placental abnormalities and congenital malformations, is also essential to reduce fetal mortality.

Abbreviations

AIC	Akaike information criterion
BIC	Bayesian information criterion
BMI	Body Mass Index
CETESB	Environmental Agency of São Paulo
ELISA	Enzyme-linked immunosorbent assay
FGR	Fetal growth restriction
FVM	Fetal Vascular Malperfusion
GA	Gestational age
HCFMUSP	Hospital das Clínicas, Faculty of Medicine, University of São Paulo
HDI	Human Development Index
ICD-11	International Classification of Diseases 11th Revision
LMP	Last Menstrual Period
MRSP	Metropolitan Region of São Paulo
MSP	Municipality of São Paulo
MVM	Maternal Vascular Malperfusion
OR	Odds Ratio
RT-PCR	Reverse transcription polymerase chain reaction
SCRN	Stillbirth Collaborative Research Network
SGA	Small for gestational age
SUS	Brazilian Unified Health System
UK	United Kingdom
U. S.	United States
US	Ultrasound
WHO	World Health Organization

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12884-025-08427-w.

Supplementary Material 1.

Acknowledgements

We would like to express our gratitude to all the participating hospitals and their dedicated teams, whose collaboration and commitment were essential to the completion of the FetRiskS study. We extend our heartfelt thanks to the mothers who generously agreed to participate — your trust and contribution made this work possible.

Authors' contributions

LJPM, GPA, DEE, ZPS, HMDN, RPVF, MFA, and NG contributed to study conceptualization. NG, MFA, RPVF, and HMDN contributed to funding acquisition. LJPM, GPA, DEE, ZPS, GFD, MSH, OAS, RS, GMF, CMM, LRM, RMZ, AJGJ, RJB, HMDN, RPVF, and MFA contributed to data curation. RS, CMM, LRM, RMZ, AJGJ, GMF, and MSH contributed to sample processing. LJPM, GPA, DEE, ZPS, GFD, MSH, LCR, CMM, LRM, RMZ, AJGJ, RPVF, RJB, RS, HMDN, MFA, and NG contributed to data analysis. LJPM, GPA, and DEE contributed to the formal analysis. LJPM, GPA, and MFA contributed to the writing of the original draft. All authors have seen and reviewed the final text and agreed with the decision to submit the manuscript for publication.

Funding
The FetRisks project is supported by the São Paulo Research Foundation (FAPESP) through grant #2016/07765-0, which also supports post-doctoral scholarships (LJPM grant #2022/13271-1; DEE grant #2023/17594-2; OAS grant #2020/05458-9; LDMP grant #2018/19592-9; RJB grant #2022/16153-0). This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001 (master student scholarship program of GFD).

Data availability
The data supporting this study's findings are available upon request from the corresponding author.

Declarations

Ethics approval and consent to participate
The study was conducted in accordance with the ethical principles of the Declaration of Helsinki. Ethical approval was obtained from the Ethics Committee of the Municipal Health Department (process n° 16509319.0.3012.5551) and from the Ethics Committee of the Hospital das Clínicas, Faculty of Medicine, University of São Paulo (process n° 16509319.0.0000.0068). Informed consent was obtained from all pregnant women.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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Received: 2 April 2025 / Accepted: 24 October 2025
Published online: 02 December 2025

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